Toward Quantum Computation for Quantum Field Theory: A Review of the Simulation of Quantum Field Dynamics

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Abstract

The Standard Model relies on Quantum Field Theory (QFT) to describe the fundamental interactions of matter. While perturbative methods have achieved remarkable success, non-perturbative phenomena, such as real-time scattering, hadronization and false-vacuum decay, remain computationally intractable for classical systems due to the sign problem and exponential growth of compute: the Entanglement Barrier. This review examines the transition from classical to quantum simulation of real-time QFT dynamics in 1+1D models, surveying recent NISQ-era results spanning emergent lattice gauge theory in Rydberg arrays, false-vacuum decay, and inelastic scattering on digital quantum computers, and assesses the present extent of quantum advantage, finding that the boundary is not fixed but a continual contest between the latest quantum hardware and ever more specialised classical algorithms.

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Acknowledgement

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1 Introduction

Quantum Field Theory (QFT) provides our most successful framework for describing the fundamental interactions of nature, from the Standard Model to critical phenomena in condensed matter. However, exact solutions of QFT are often non-analytic, necessitating numerical simulation to explore these exotic dynamics. Classical computing methods have provided deep insight into QFT but we are reaching the limits of current hardware [1], and even sophisticated recent algorithms [2] face strict physical constraints on their applicability.

The most severe constraint emerges in the dynamical regime. While classical algorithms excel at calculating static and equilibrium properties, simulating real-time, non-equilibrium dynamics leads to an exponential explosion in required computational resources. This intractability is driven by fundamental phenomena such as the 'numerical sign problem' in path integrals [3] and the 'entanglement barrier' in tensor networks [4]. To circumvent these barriers, the field is turning toward Richard Feynman's original 1982 proposition [5]: simulating quantum mechanics directly on quantum hardware.

This review examines the current state of this transition from classical to quantum simulation, evaluating the extent of 'quantum advantage' in QFT simulations. To probe the capabilities of current quantum devices against classical limits, this review will largely consider $1+1$ dimensional (1 space + 1 time) models, such as the Ising and the Schwinger model. Despite their reduced dimensionality, these models capture a richness of high-energy phenomena; from inelastic scattering and particle production to vacuum decay and meson confinement, serving as ideal, near-term test-beds.

We proceed from theoretical preliminaries for the Ising and Schwinger models, through classical benchmarks, to both analogue and digital quantum computational approaches, before assessing the state of 'quantum advantage'.

2 Theoretical Framework

2.1 Ising Model

The general Quantum Ising model is a simple $1+1D$ lattice quantum spin model described by the Hamiltonian [6]

$$\hat{H} = -J \sum_i \hat{\sigma}_i^z \, \hat{\sigma}_{i+1}^z - h_x \sum_i \hat{\sigma}_i^x - h_z \sum_i \hat{\sigma}_i^z \quad (1)$$

J is the nearest neighbour coupling between spins represented by Pauli matrices $\hat{\sigma}^z$. The second term describes the external transverse magnetic field with a field strength, h_x. When $h_z = 0$ eq.1 becomes that of the Transverse Field Ising Model (TFIM). In the case of $h_z \neq 0$, the $\mathbb{Z}_2$ symmetry of the model is explicitly broken.

The difference between eq.1 and the classical Ising model is the non-commutation of $\hat{\sigma}^x$ and $\hat{\sigma}^z$, meaning the transverse and longitudinal spin projections have no simultaneous eigenstates. Unlike the classical Ising model, TFIM does possess a critical phase transition at the point $h_x = J$ at which the correlation length, $\xi \to \infty$.

2.2 Schwinger Model

The Schwinger model is a $1+1$ dimensional model of Quantum Electrodynamics (QED) with a Dirac fermion field, ψ, and Lagrangian density:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - e\gamma^\mu A_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (2)$$

Where $\bar{\psi} \equiv \psi^\dagger \gamma^0$(the Dirac Adjoint), γ^μ the Dirac gamma matrices, and $F^{\mu\nu}$ is the Electromagnetic field strength tensor. This is a continuum field theory and needs to be discretised onto a lattice for computations.

Naively putting the fermion fields on a lattice produces twice as many fermion states as expected, in 1-spatial dimension. The commonly used Kogut-Susskind formulation [7] avoids this 'fermion-doubling problem' by staggering the two components of the Dirac spinor onto ad-

jacent lattice sites such that for ψ and a one-component fermion field $\phi(n)$,

$$\psi = \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix}, \qquad \begin{aligned} \psi_e(n) &= \phi(n), \quad n \text{ even} \\ \psi_o(n) &= \phi(n), \quad n \text{ odd.} \end{aligned} \tag{3}$$

2.2.1 Jordan-Wigner Transformation and the continuum limit

To simulate the Schwinger model classically, or indeed on a quantum computer, the fermion operators ($\phi(n)$) must be written in terms of Pauli operators using the Jordan-Wigner Transformation [6]:

$$\begin{aligned} \hat{\phi}_i &= (\prod_{k<i} \hat{\sigma}_k^x) \frac{\hat{\sigma}_i^z - i\hat{\sigma}_i^y}{2} \\ \hat{\phi}_i^\dagger &= (\prod_{k<i} \hat{\sigma}_k^x) \frac{\hat{\sigma}_i^z + i\hat{\sigma}_i^y}{2} \end{aligned} \tag{4}$$

We thus have a mapping of fermionic degrees of freedom to the spin chain (Pauli operators), suitable for simulation with MPS or quantum computing.

3 Classical Benchmark

Several classical methods complement MPS for QFT simulation. Monte-Carlo path-integral methods [8] are highly effective for static and equilibrium properties, but in real time, the action's complex phase factor prevents importance sampling; the sign problem. Form-Factor perturbation theory (FFPT) [4] gives analytic insight near integrable limits but diverges for strongly-coupled regimes. Truncated Hamiltonian methods [9] give non-perturbative energy spectra by diagonalising in a finite Hilbert subspace; however, similar to MC methods, these do not access real-time dynamics.

Matrix Product States are a class of tensor network (TN) ansatz that represent a quantum many-body state as a contracted chain of, usually, rank-3 tensors [10] and have emerged as one of the most capable methods for classical simulation of real-time 1+1-dimensional quantum dynamics. Ising field theory (IFT) is a useful first demonstration of the capabilities of MPS methods, owing to its integrable and non-integrable limits, which enable comparison with known analytic values and exploration of strongly coupled regimes. To parametrise the trajectory between the two integrable limits, one can define the dimensionless ratio of the magnetic field couplings:

$$\eta_{latt} = \frac{h_x - 1}{|h_z|^{\frac{8}{15}}} \tag{5}$$

in units where $J = 1$ [4]. The two integrable limits of IFT are the Massive Free Fermion theory at $\eta \to \infty\,(h_z = 0)$, corresponding to the continuum limit of the Ising model, in which scattering is trivial; and Zamolodchikov's Massive E_8 theory at $\eta = 0\,(h_x = 1)$, known to host eight stable particles.

3.1 How far can MPS simulations go?

Initial MPS simulations align well with known FFPT calculations (See Appendix A) near the integrable limits, but how well do they work in non-integrable regimes for simulating real-time dynamics?

In the exact E_8 limit, the heavy m_4 particle is stable but acquires a finite decay width, Γ_4, when perturbed away from $\eta = 0$. The authors simulated the collision of two m_1 particles at a centre-of-mass energy $= m_4$, projected the post-collision wavefunction onto the $|1,1\rangle$ MPS basis (two separate m_1 particles), and computed the probability of not being in this state at time, t, ($P(\neg 11, t)$). The decay width of the m_4 resonance could thus be found by fitting the measured $P(\neg 11, t)$ with

$$P(\neg 11, t) = Ae^{-\Gamma_4 t} + P_{leak} \tag{6}$$

P_{leak} accounts for stable particle production ($m_1 + m_1 \to m_1 + m_2$) i.e. a leakage into inelastic channels.

One notices in Fig.1 that while MPS results align well with Truncated Hamiltonian calculations, there is a discrepancy far from the E_8 limit. The authors of ref.[4] note that this is due to the increased leakage into inelas-

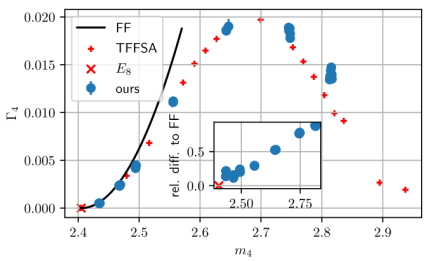


Figure 1: [4] Decay width Γ_4 against mass m_4, obtained from fitting to their simulations (blue dots) with the form-factor prediction (black line) and TFFSA data (red crosses) taken from [12]. Inset: Relative difference between simulated and form-factor values, showing convergence as we approach the E_8 point.

tic channels but could be resolved by simulating on a greater number of sites (N), reducing the wave-packet width in energy-momentum space. While this invokes a computational cost, complexity is $\mathcal{O}(N)$ [11] and so is efficient. This is not a limit for MPS simulations.

3.1.1 Strongly-Coupled dynamics

MPS proves an excellent capacity for 'dynamical spectroscopy' in real-time high-energy regimes. There exists a conjecture by Zamolodchikov [13]; that for high-energy scattering events in the free-fermion or E_8 limit, scattering is dominated by inelastic/elastic scattering, respectively. MPS simulations to confirm this are non-trivial due to significant lattice artefacts at high energies. Thus, one requires a small ratio of lattice spacing to correlation length, a computationally expensive requirement. Despite this, Fig.2 from Jha et al. (2024) supports the Zamolodchikov conjecture.

Similarly, MPS methods allow the observation of the momentum threshold for particle production in the lattice Schwinger Model for meson-collisions in ref.[14]. Measurement of the two-site entanglement entropy provides clear visuals for the real-time scattering in the model (Fig.3), indicative of the 'dynamic spectroscopy' that tensor-network methods provide.

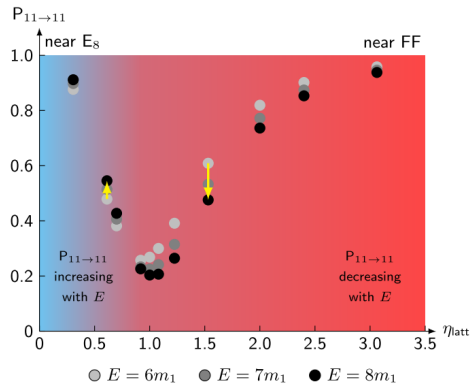


Figure 2: The high energy behaviour of scattering in IFT for E = 6, 7, 8 in units of the lightest particle mass. The results indicate that the large-E behaviour changes around $0.7 \lessapprox \eta_{c,latt} \lessapprox 0.9$. From Jha et al. (2024)[4]

3.2 The Entanglement Barrier

While TN methods have successfully produced simulations of some strongly-coupled regimes in Ising and Schwinger models; their universal applicability in QFT simulations is limited by the computational resources necessary to maintain accuracy in high-energy, high-entanglement events as well as for longer time simulations. In MPS simulations, the entanglement entropy, S, of an event is known to scale linearly with time. The bond-dimension, χ (the matrix size in the tensor network), scales exponentially with S [4, 15, 14, 16]. Thus χ scales exponentially with time, and therefore the computational cost of the simulation. Highly entangled scattering where particle production is abundant remains, in general, computationally inaccessible [4, 14].

4 Quantum Computation

The Quantum Church-Turing thesis posits that a quantum computer will be capable of efficiently simulating any process that occurs in nature [17], a solution to the limits of classical computation. Quantum simulations proceed along two complementary tracks: digital gate-based computers, targeting universality, and analogue simulators.

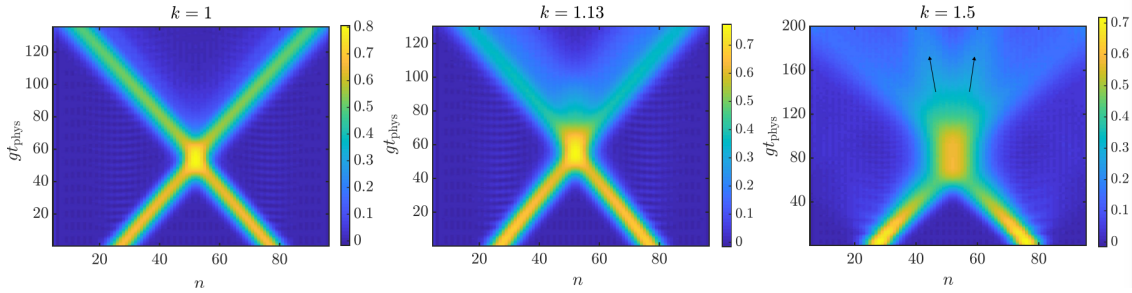


Figure 3: Real-time meson scattering in the lattice Schwinger model at three incoming momenta. Left ($k = 1$): only the elastic channel is open. Middle ($k = 1.13$): just above the inelastic threshold predicted by variational MPS, two bound states are produced with near-zero velocity, visible below $\text{gt}_{\text{phys}} \sim 80$. Right ($k = 1.5$): both channels open; the inelastically produced bound states have opposite, smaller velocities than the elastic outgoing mesons. From Papaefstathiou et al. (2025)[14]

4.1 Analogue Quantum Simulation

Analogue quantum devices are tunable physical systems whose 'native' Hamiltonian, H_{sim}, can be mapped onto the target Hamiltonian, H_{target}, of the QFT. In this way, analogue devices are able to evolve continuously in time under H_{target}. The general consensus, however, is that such devices are only an ephemeral technology during the Noisy Intermediate-Scale Quantum (NISQ) era until fault-tolerant quantum computers arise [17].

4.1.1 False Vacuum Decay

Lagnese et al. (2021) [18] demonstrated that false vacuum decay can be mapped onto an Ising spin chain (with fields h_x and h_z) by preparing the system in the fully aligned state ($\sigma_z = 1$ on every site), and evolving the system in imaginary time using iTEBD to converge on the ground state and finally quenching $-h_z \to h_z$. False vacuum decay was then observed using real-time evolution with TN methods.

False vacuum decay occurs via the nucleation and rapid expansion of 'bubbles' of the true vacuum with the decay rate of false-vacuum per lattice site [18]:

$$\gamma \propto h_z \, e^{-\frac{q}{h_z}} \tag{7}$$

Where $q = q(h_x)$ is essentially parametrising the energy cost of nucleating a bubble of true vacuum. Hence, for small but non-zero h_z the false-vacuum decay time (γ^{-1}) becomes intractably large for classical computers (the entanglement barrier again) and so the authors suggest that quantum simulators, such as trapped ions or Rydberg atoms, could be used to simulate these long-time regimes with cosmological analogues. Crucially, they discuss how to modify an existing trapped-ion quantum simulator that successfully simulated confinement in many-body systems [19].

4.1.2 Emergent Lattice gauge theories

In a demonstration of the further utility of analogue quantum simulations, a 2017 paper by Bernien et al. [20] observed unexpected non-thermalising oscillations (see figure 4) upon quenching of the 'Ising-like' PXP Hamiltonian, which the authors attribute to the 'constrained dynamics due to Rydberg blockade'.

It was on these observations that Surace et al. [21] proposed that the restricted kinematics of the Rydberg array map exactly onto $U(1)$ Lattice Gauge Theory (LGT). Specifically, the authors suggest that the Rydberg blockade in ref.[20] is a physical counterpart to Gauss' Law in gauge Theories, constraining the Hilbert space in the same manner. Under this exact mapping, the unexpected non-thermalising oscillations observed in ref.[20] are reinterpreted as 'string inversion'. This phenomenon is directly tied to string breaking (i.e. Schwinger pair production) typical of gauge theories featuring dynamical matter [21, 22].

The results of ref.[20] and ref.[21] demonstrate the

physical simulation of real-time lattice gauge theory dynamics; a formidable classical computation. In the NISQ era, they have shown that analogue quantum simulators are a powerful intermediary able to exploit the natural constraints of physical systems.

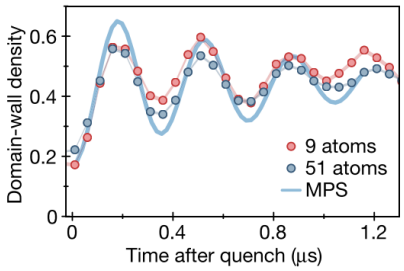


Figure 4: Domain-wall density after the quench. The dynamics decay slowly on a timescale of 0.88 μs. Shaded region represents the standard error of the mean. Solid blue line is a fully coherent matrix product state (MPS) simulation with bond dimension D = 256, taking into account measurement fidelity. [20]

4.2 Digital Quantum Computation

While Analogue quantum simulators offer an excellent platform for simulation of specific Hamiltonians, we cannot simulate all general interactions on an analogue device [2] due to difficulty physically realising the Hamiltonians and susceptibility to calibration errors [23].

Digital quantum computing provides the pathway to universality. By decomposing time evolution into discrete gates operating on qubits encoding the field, digital systems can, in principle, simulate any quantum field theory. Current digital simulations have moved beyond proof-of-principle demonstrations and into the NISQ (Noisy Intermediate-Scale Quantum) era [17], allowing observations of distinctively quantum-field-theoretic phenomena.

4.2.1 Inelasticity in Ising Field Theory

A landmark result was achieved by Farrell et al. (2025) [16] using 104 superconducting qubits on the IBM *Marrakesh* processor to simulate particle production through inelastic scattering in the $1+1D$ Ising model; the first time particle production was observed in this manner.

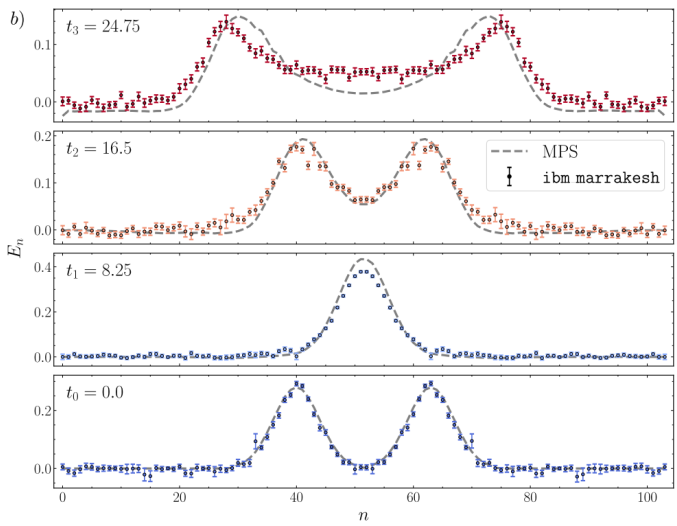


Figure 5: Results from simulations of scattering using L = 104 qubits of IBM *Marrakesh* for a selection of times. The wavepackets are initialised at $t_0 = 0$ and collide around $t_1 = 8.25$. Outgoing particles begin to form at $t_2 = 16.5$ and $t_3 = 24.75$. The asymmetry of the energy density in each "bump" signals the formation of the $|2\rangle$ particle produced in the inelastic process $11 \to 12$. [16]

Two ingoing wavepackets with centre of mass energy, $E_{cm} \geq m_1 + m_2$, were initialised, opening up the lowest energy inelastic process $11 \to 12$.

The authors were able to 'observe' inelastic particle production via the skewness of the outgoing energy density (see Fig.5). Using $5,589$ 2-qubit gates, an inelastic scattering event was simulated for time t=24.75, after which the cumulative noise rendered simulations impossible. Because the asymptotic particle tracks did not have sufficient time to separate (and thus make clear observations), outgoing energy density measurements are a useful NISQ method.

Critical to this result was Farrell et al.'s departure from adiabatic state preparation. Their W-state-based initialisation protocol [16] employed mid-circuit measurements and feed-forward operations to construct the initial state in constant circuit depth, $\mathcal{O}(1)$; preserving the processor's coherence budget for the highly entangled collision dynamics.

4.2.2 Fermionic and Meson Scattering in the Schwinger Model

While Ising spin models are a useful test-bed of techniques, we need a more intricate model to explore richer dynamics. Schuhmacher et al. (2025) [24] utilise IBM *Marrakesh*, but with 1+1D lattice gauge theory: the Schwinger Model. While this is a theory of Quantum Electrodynamics (QED), its formulation in 1+1D exhibits features common with Quantum Chromodynamics (QCD) [14], like confinement, making it an excellent model for a variety of QFT dynamics.

The authors of ref.[24] apply a confining potential to electron-positron[1] collisions and observe the propagation of the confined pair post-collision. More interestingly, it is demonstrated that by quenching the electron mass at the moment of collision (equivalent to a high-energy relativistic collision), three distinct regimes occur depending on the final quenched mass of the electron, m_f. The authors initialise the collisions with an electron mass of $m_i = 1.5\kappa$ and zero confining potential.

- $\mathbf{m_f = 0\ \kappa}$ **:**
  Inelastic scattering was observed, with persistent meson production and annihilation, which lead to string inversions between the electron and positron, the same phenomenon as observed in Bernien et al. [20] and discussed in section 4.1.2.

- $\mathbf{m_f = m_c = 0.3275\ \kappa}$ **:**
  At this critical point, near-complete annihilation of matter is observed as the system relaxes to the vacuum state. This critical quenched mass represents an interesting phase transition in the 1+1D Schwinger model, which we will discuss further.

- $\mathbf{m_f = 0.8\ \kappa}$ **:**
  When the mass was quenched to greater than the critical mass, elastic scattering was recovered, displaying, qualitatively, the same dynamics as non-quenched collisions.

$m_f = 0.3275\kappa$ represents a so-called 'Coleman's phase transition' [25], a second-order transition related to the mass-dependent spontaneous breaking of Charge-Parity (CP) symmetry [24] at zero confining potential in the Schwinger Model. Beyond intrinsic interest, it is exploited [24, 26] to probe high-energy collisions in quantum field theory without requiring exceptionally deep circuit depths and accumulating significant noise before the collision.

The major leap made by Schuhmacher et al. is simulating the collision of composite particles, mesons. Post-collision dynamics on IBM *Marrakesh* are noisy and do not yet improve on MPS solutions, but the proof of concept is a crucial step towards future simulations.

5 The extent of quantum advantage?

Quantum computers are growing ever more capable, but how close are we to demonstrating 'advantage'? A 2023 paper by Kim et al. [27] aimed to demonstrate the utility of IBM's 127-qubit Eagle processor in the pre-fault-tolerant era on the Trotterised time evolution of a 2D transverse-field Ising model. With Zero-Noise Extrapolation, Eagle produced reliable expectation values (see Fig.6) in regimes where general-purpose classical methods (1D MPS and 2D isometric TN states) broke down under the entanglement growth.

An obvious issue with this result is the use of 'general-purpose' TN methods rather than a specialised network; indeed, a later paper by Tindall et al. [28] implemented a TNS mirroring the 'heavy-hexagon' geometry of IBM Eagle. Due to the 'treelike' entanglement generated by this circuit geometry, the authors utilised belief propagation, which excels on these sparse networks. Tindall's classical approach produced a magnetisation to an accuracy of 10^{-14} running in less than 10 seconds on a MacBook Pro. Kim et al. [27] demonstrated impressive error mitigation on NISQ hardware, but the classical benchmark itself continues to move.

[1] We note for clarity that while Schuhmacher [24] refers to the particles in the Schwinger Model as electrons/positrons, they are just the fundamental fermions of the model [25] elsewhere referred to as 'quarks'.

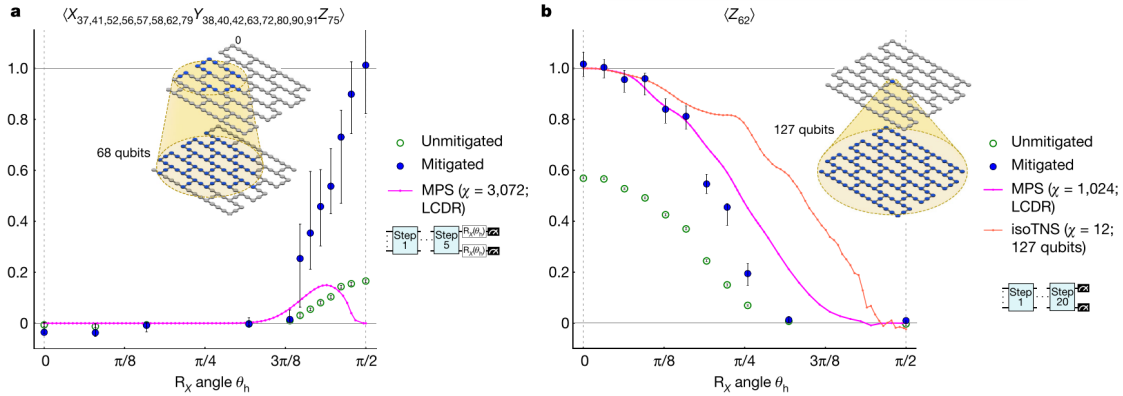


Figure 6: **Classical methods at their breaking point in the IBM 127-qubit kicked Ising experiment**. (a) A weight-17 observable after five Trotter steps. Even with light-cone optimisation and bond dimension $\chi = 3,072$, MPS fails to reproduce the exact stabiliser value at $\theta_h = \frac{\pi}{2}$ (y-axis negated). (b) Single-site magnetisation $\langle Z_{62} \rangle$ after 20 Trotter steps; the light-cone-optimised MPS ($\chi = 1,024$) and isoTNS ($\chi = 12$) both diverge from the error-mitigated quantum estimates. From Kim et al. [27](Fig.4)

6 Conclusion and outlook

This review has traced the transition from classical to quantum simulation of real-time QFT dynamics in 1+1D models. Tensor network methods, exemplified by MPS, remain the classical benchmark and have produced significant results (Jha et al.'s dynamical spectroscopy near the E_8 point, Papaefstathiou et al.'s observation of the inelastic threshold in meson scattering) but are fundamentally constrained by the linear growth of entanglement and the corresponding exponential cost in bond dimension.

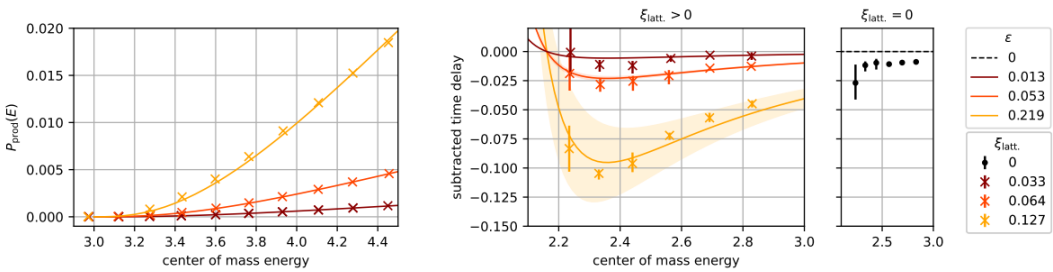
Quantum platforms are beginning to reach regimes the entanglement barrier has made difficult to reach classically. Analogue simulators have demonstrated emergent $U(1)$ lattice gauge theory in Rydberg arrays, and proposed implementations of false-vacuum decay on existing trapped-ion hardware. Digitally, Farrell et al. observed inelastic particle production on superconducting qubits using their constant-depth W-state preparation protocol, and Schuhmacher et al. simulated meson collisions in the Schwinger model, the first hadron scattering results on a quantum computer.

The results of Schuhmacher and Farrell do not demonstrate quantum advantage, but represent an important proof-of-principle for future fault-tolerant particle collision simulators. The Kim-Tindall papers demonstrated that quantum hardware claims benchmarked only against general-purpose classical methods can be undone by more specialised algorithms exploiting particular properties of the system. Quantum advantage is less a threshold to cross than a horizon that recedes; as quantum hardware advances, so too do the classical algorithms it provokes. Near-term progress towards quantum advantage could likely come from QFT simulations where classical simulations are not able to exploit underlying structure. With future fault tolerance, the techniques demonstrated on these 1+1D testbeds could become tools of discovery in 3+1D QFT simulation.

7 Appendices

A MPS-FFPT comparisons near integrable limits

Figures (source Jha et al. [4]) for the initial demonstration of MPS accuracy near the two integrable limits compared to known FFPT calculations.



(a) (b)

Figure 7: (a) The simulated production probability at three different values of the lattice coupling parameter ξ_{latt}. The solid lines are single-parameter fits of the theoretical production probability at first order in FF perturbation theory. (b) The simulated time delay at the same values of ξ_{latt}. The theoretical predictions from FF perturbation theory are completely fixed by the probability fits. The shaded regions show the expected magnitude of the second-order correction.

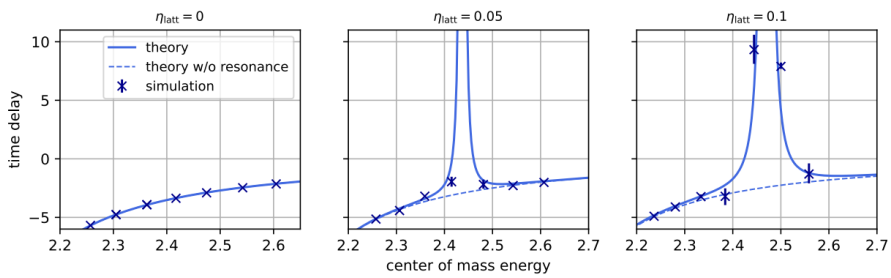


Figure 8: Time delay for elastic $11 \to 11$ scattering near E_8. The solid lines show the predictions from form factor perturbation theory. The dashed lines show the expected time delay in the absence of resonances.[4]

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